Aprendizagem Computacional
Machine Learning

2026/2027

Project Assignment

Diabetes Risk Prediction

1 Background

Diabetes is a chronic disease that develops when the pancreas does not produce enough insulin or when the body cannot effectively use the insulin it produces. This leads to hyperglycaemia, also known as high blood sugar, which, if left unmanaged, can cause serious damage to many parts of the body, including the heart, eyes, nerves, blood vessels and kidneys. This disease affects around 830 million people worldwide, a number that has more than quadrupled since 1990. [1, 2]

2 Objective

Your task is to develop machine learning pipelines to predict if a person is at a low, medium or high risk of developing a diabetes condition.

3 Practical Assignment

The report is expected to have a structure similar to this section up until subsection 3.5.

3.1 Dataset Description

Consider the synthetic dataset available at
https://www.kaggle.com/datasets/mansiaggarwal88/diabetes-risk-prediction that contains 17 demographic, lifestyle, and clinical features for 15 000 patients. You should perform an exploratory analysis of the dataset before moving on with other tasks.

3.2 Data partitioning

Implement a training, validation and testing data partitioning approach to perform a trustworthy performance assessment of your pipelines.

3.3 Feature Selection and Reduction

Some of the supplied features may be useless, redundant, or highly correlated with others. Consider using feature selection (Kruskal-Wallis, ROC-AUC and correlation analysis) and dimensionality reduction (PCA

and LDA) techniques to evaluate their impact on the performance. Analyse the distribution of your feature values and calculate the correlation between them. Ensure you understand your features! Remember to include your findings in the final report.

3.4 Classification

Implement just the classifiers studied during the course. Other classifiers are not important. Thus, you should consider the classifiers: Minimum Distance, Fisher LDA, Bayes classifier (not Naive Bayes), kNN, SVM (Linear+Non-Linear), AdaBoost, Random Forests and Decision Trees.

3.5 Performance Assessment & Causality Analysis

Consider the performance metrics learned during the course to report the performance of your pipelines, such as: Sensitivity, Specificity, Precision, F1 score, and ROC-AUC.

Run the experiments multiple times and present average/median results and standard deviations /IQR(for the metrics used).

Remember that manually inspecting your algorithm's predictions can provide valuable insights into where it is failing and why, as well as how to improve it (e.g., what causes the algorithm to fail in specific cases? What unique characteristic makes it particularly challenging? How can I help the algorithm handle those cases more effectively?). Reassess the Pre-processing, Feature Reduction, and Feature Selection phases until you are satisfied with the results. It's wise to monitor the evolution of your algorithm's performance throughout this process. Aim to display these trends in your final report to support all related issues (parameter selection, model fit, etc.).

You can write your own code or utilise the available functions and methods. The methods employed in your work should be described and understood by you. Be aware of what is the impact of the selected parameters.

Present and discuss the results obtained from your project assignment.

3.6 Submissions

You have to submit a report (in Portuguese or English) in PDF format and the developed code.

⇒ **Deadline: December 18, 2026!**

3.7 Requirements

⇒ The practical assignment should be developed in pairs. Larger groups are not permitted!

⇒ The submitted report is limited to **10 pages (single-column).**

⇒ The usage of LaTeX and of the Springer LNCS Template is encouraged.

⇒ Any usage of Generative Artificial Intelligence in the development of this project must be disclosed in both the report and code.

References

[1] World Health Organization, "Diabetes", 2024, https://www.who.int/news-room/fact-sheets/detail/diabetes.

[2] World Health Organisation, "WHO Global Diabetes Compact", 2024, https://www.who.int/initiatives/the-who-global-diabetes-compact